

Machine Learning

Time: 3 Hours

Max Marks: 75

SECTION – A

(5 × 5 = 25 Marks)

Answer any FIVE of the following. Each question carries 5 marks.

1. Define Machine Learning. Write its applications.
2. Distinguish between Supervised and Unsupervised Learning.
3. Explain the steps involved in K-Means Clustering.
4. What is Logistic Regression?
5. Define Confusion Matrix and explain its components.
6. Write a short note on Overfitting and Underfitting.
7. What is Principal Component Analysis (PCA)?
8. List any five Python libraries used in Machine Learning.

SECTION – B

(5 × 10 = 50 Marks)

Answer ALL the following questions. Each question carries 10 marks.

- 9 . (a) Define Machine Learning. Explain its types with suitable examples.

OR

- (b) Describe the components of a Machine Learning system and the ML workflow.

10. (a) Explain Linear Regression with mathematical formula and Python implementation.

OR

- (b) Explain the working of K-Nearest Neighbors (KNN) algorithm with an example.

- 11 . (a) Describe the working of K-Means Clustering with a suitable diagram.

OR

- (b) Explain DBSCAN algorithm and its advantages over K-Means.

12. (a) Define and explain Precision, Recall, F1-score with example.

OR

- (b) Explain Bias-Variance Tradeoff and the concept of Cross-validation.

UNIT – V

13. (a) Write a Python program using Scikit-learn to implement Logistic Regression on a dataset.

OR

- (b) Explain the use of preprocessing in ML. Write a Python code for scaling and encoding data.